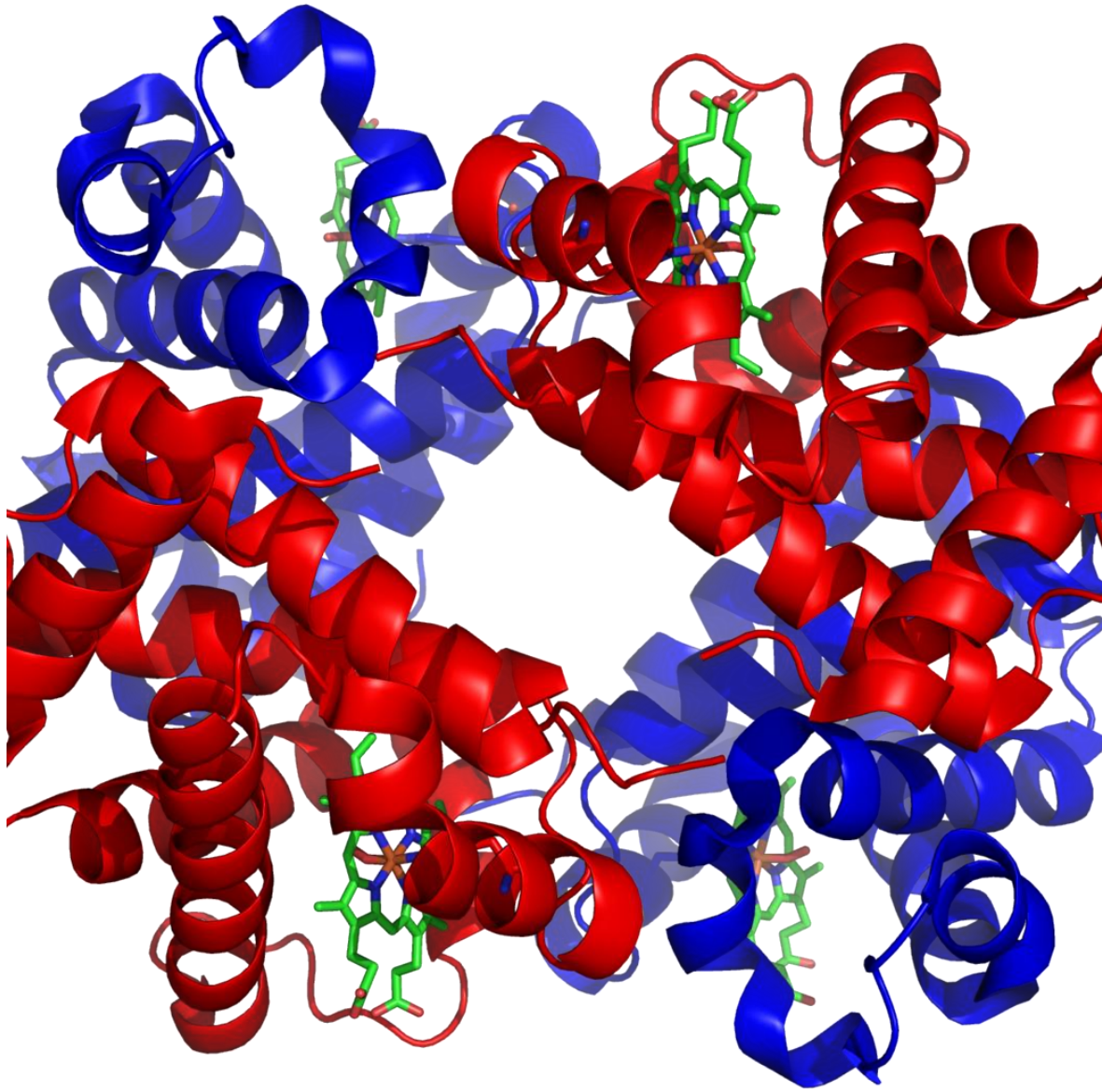
DP Biology – B1.2 (HL extension)

Protein structure

- How does the order of amino acids produce such a variety of protein forms and functions?
- How do chemical and physical conditions affect protein molecules?



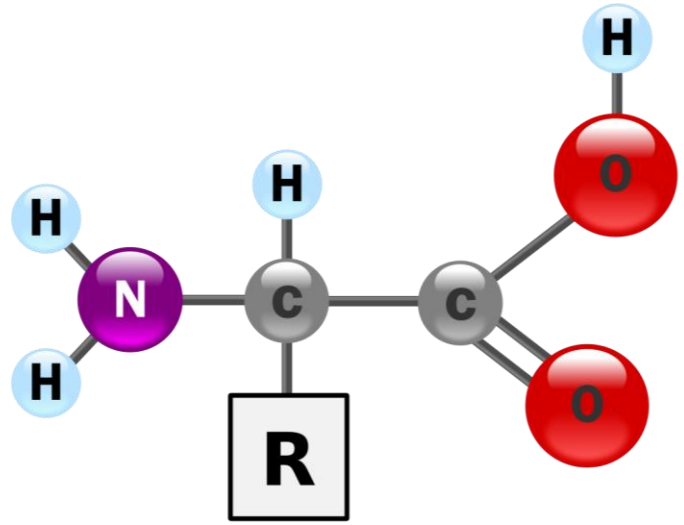
HL statements B1.2.6–12. Textbook pp. 210–222.
Image: haemoglobin, four chains with haem groups (green). Richard Wheeler, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	B1.2.6–8	Explain how R-groups give proteins their properties, and how primary structure leads to secondary structure
2	B1.2.9–10	Explain the bonds that hold tertiary structure, and where hydrophobic and hydrophilic amino acids end up
3	B1.2.11–12	Compare quaternary structure in insulin, collagen and haemoglobin, and relate globular and fibrous forms to function

R-groups make the difference

B1.2.6



General structure of an amino acid. Public domain.

Type of R-group	Properties
Non-polar	Hydrophobic: does not form hydrogen bonds with water
Polar, uncharged	Hydrophilic: forms hydrogen bonds
Acidic	Extra -COOH that loses H^+ to become negative
Basic	Extra amine group that gains H^+ to become positive

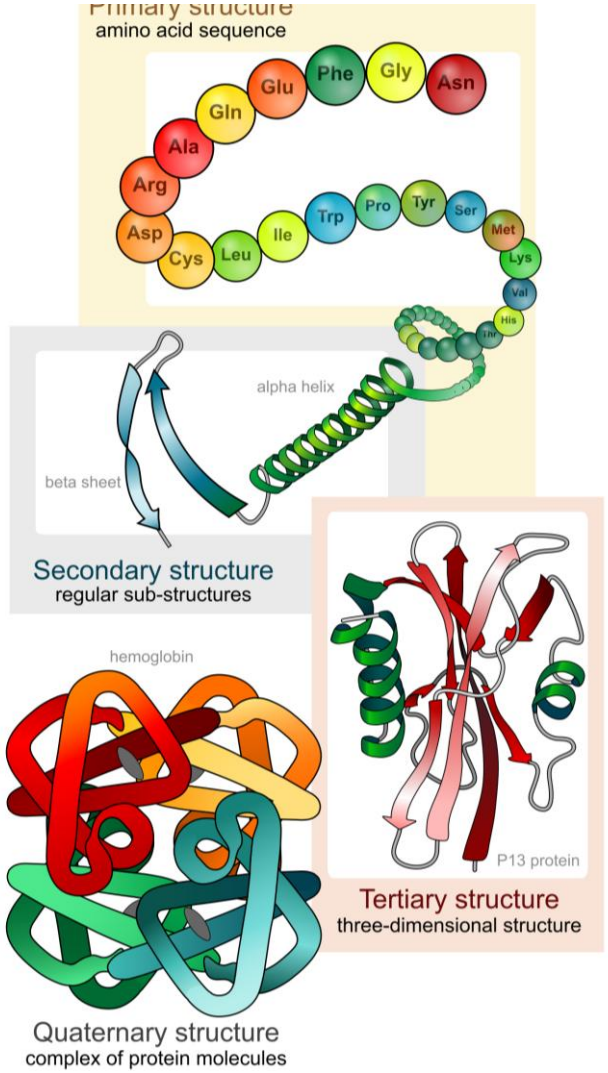
All 20 amino acids share the same backbone, so only the **R-group** differs. R-groups vary in size, charge and polarity, and some (cysteine) can form covalent bonds with each other.

Key idea

Twenty different R-groups in any order give almost unlimited sequences. The R-groups decide how a chain folds and what it can bind.

Primary structure: the sequence decides the shape

B1.2.7



Levels of protein structure. LadyofHats, public domain.

The **primary structure** is the sequence of amino acids in a polypeptide, set by the base sequence of its gene.

The sequence fixes where each R-group sits along the chain, and therefore which R-groups can interact. So it determines how the chain folds: every copy of a protein has the same precise shape.

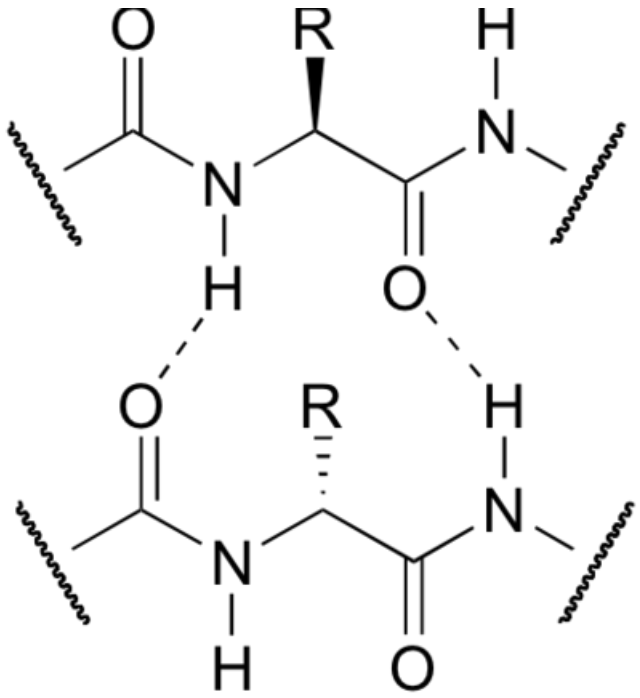
Small proteins that have been unfolded can often refold correctly on their own, which shows that the information for folding is in the sequence.

Think about it

Why can a computer predict a protein's shape from its sequence, but not from a list of how many of each amino acid it contains?

Secondary structure: helices and pleated sheets

B1.2.8



hydrogen bonding in a β-strand

Hydrogen bonds between C=O and N-H in the backbone. CC BY-SA

	Alpha helix	Beta-pleated sheet
Shape	Right-handed coil	Strands lying side by side, pleated
Hydrogen bonds	C=O to N-H four amino acids along the same chain	C=O to N-H of neighbouring strands
Pattern	Regular	Regular

Common mistake

Secondary structure is held by hydrogen bonds between atoms of the backbone (C=O and N-H), not between R-groups. R-group bonds belong to tertiary structure.

Tertiary structure: bonds between R-groups

B1.2.9

Bond	Between	Strength and sensitivity
Hydrogen bonds	Polar R-groups	Weak; broken by heat
Ionic bonds	Positively charged amine and negatively charged carboxyl R-groups	Moderate; changed by pH
Disulfide bonds	The -SH groups of two cysteines	Strong covalent bond
Hydrophobic interactions	Non-polar R-groups clustering away from water	Weak individually; many together

Charge depends on pH

Amine groups gain H^+ and carboxyl groups lose H^+, so whether ionic bonds form depends on the pH. This is why pH can denature proteins.

Hydrophobic inside, hydrophilic outside

B1.2.10

Protein type	Where amino acids go
Soluble globular protein	Non-polar in the core; polar on the surface, in contact with water
Integral membrane protein	Non-polar where it crosses the lipid bilayer; polar where it meets water on either side
Channel protein	Polar amino acids also line the pore, so ions can pass

Activity

A protein has a stretch of about 20 amino acids that are all non-polar. Predict where in the cell this stretch is likely to be, and explain. Then predict what would happen if a polar amino acid replaced one in the middle.

Check-in: R-groups and folding

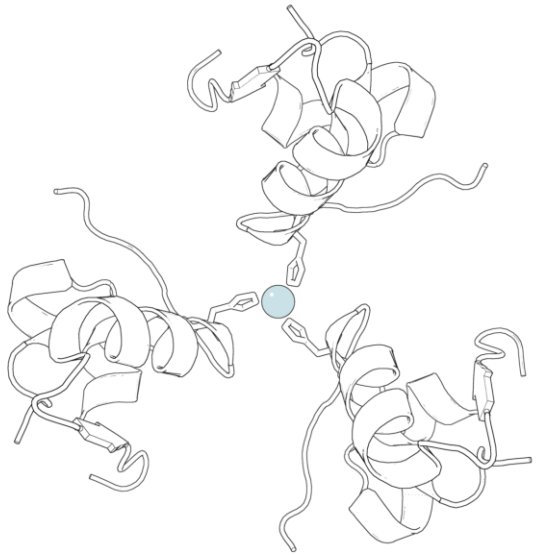
B1.2.6–10

1. Classify an R-group with an extra -COOH group.
2. Explain why the primary structure determines a protein's shape.
3. State the atoms joined by hydrogen bonds in an alpha helix.
4. List four types of bond that stabilize tertiary structure.
5. State where non-polar amino acids are found in a soluble globular protein.

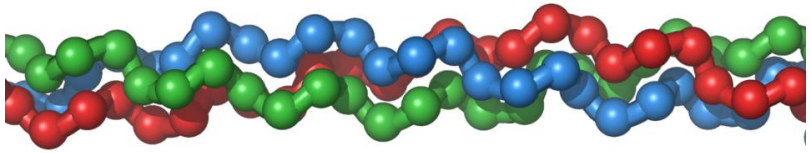
How to use this

Answer without your notes, then compare with a partner.

Quaternary structure



Insulin, stored as six molecules (hexamer). Public domain.



Collagen triple helix. CC BY-SA 3.0.

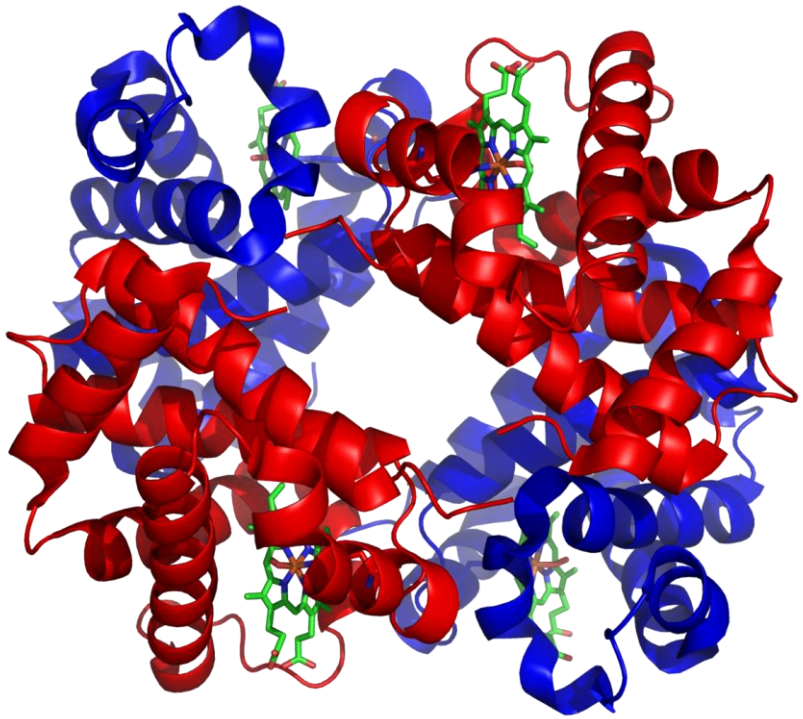
Protein	Polypeptides	Conjugated?
Insulin	Two chains (A and B) joined by disulfide bonds	No
Collagen	Three chains wound into a triple helix	No
Haemoglobin	Four chains, each with a haem group	Yes: haem is a prosthetic group

Definitions

Quaternary structure: two or more polypeptides joined together. A conjugated protein also has a non-polypeptide part, a prosthetic group, such as haem with its iron.

Haemoglobin: a conjugated protein

B1.2.11



Haemoglobin: alpha chains red, beta chains blue, haem green. CC BY-SA 3.0.

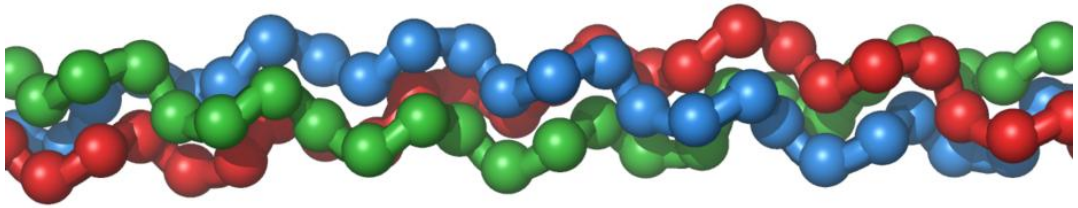
Haemoglobin has **four polypeptides**: two alpha and two beta globin chains.

Each chain holds one **haem group**, with an **iron ion** at its centre. Each iron ion binds one oxygen molecule, so one haemoglobin carries up to four.

The chains fit together so that when one oxygen binds, the shape changes slightly and the next binds more easily. This **cooperative binding** helps haemoglobin load oxygen in the lungs and unload it in respiring tissues.

Collagen: a fibrous protein

B1.2.12



Collagen triple helix. CC BY-SA 3.0.

Link
Vitamin C is needed to make collagen. Without it, collagen is weak: scurvy.

Collagen is the most abundant protein in mammals: it gives strength to skin, tendons, ligaments, cartilage and bone.

Its sequence is highly repetitive: **every third amino acid is glycine**, the smallest, which lets three chains wind tightly into a **triple helix**. Many triple helices line up and are cross-linked into fibres that are very strong in tension.

Form and function: globular and fibrous

B1.2.12

	Insulin (globular)	Collagen (fibrous)
Shape	Compact and roughly spherical	Long, rope-like triple helix
Solubility	Soluble: carried in blood plasma	Insoluble
Sequence	Varied; folds into one precise shape	Repetitive; every third amino acid is glycine
Function	Hormone: binds a specific receptor on target cells	Strength in tissues, resisting pulling forces

Key idea

Globular proteins (enzymes, hormones, antibodies) need precise shapes to bind other molecules. Fibrous proteins need length and strength.

Working with data: heat and a protein’s shape

B1.2.9

Activity of an enzyme after 10 minutes at each temperature, then measured at 25 °C. These data are invented for practice; the pattern is realistic.

Pre-heated at / °C	Activity remaining / %
25	100
40	98
50	80
60	35
70	5
80	0

1. Describe the effect of heating on the activity remaining.
2. Explain the results in terms of bonds in tertiary structure.
3. The enzyme has several disulfide bonds. Suggest why activity is still lost at 80 °C.
4. Suggest how an enzyme from a hot-spring bacterium might differ in structure.

Protein detective

B1.2.6–12

1. Open a protein structure in a 3D viewer (for example insulin or collagen on PDB-101). Colour it by hydrophobicity. Where are the hydrophobic amino acids?
2. Find two cysteines joined by a disulfide bond in insulin.
3. **Explain** why boiling denatures insulin but does not break its disulfide bonds.

Link

Denaturation, and why an enzyme's shape matters, are covered in B1.2 (SL) and C1.1.

Check-in: quaternary structure and form

B1.2.11–12

1. Define quaternary structure.
2. Explain why haemoglobin is called a conjugated protein.
3. Explain why glycine is needed at every third position in collagen.
4. Distinguish between globular and fibrous proteins, using insulin and collagen.
5. Suggest why a fibrous protein would be no use as a hormone.

How to use this

Write full sentences, then mark with a partner.

Common mistakes to avoid

Not this	But this
Secondary structure is held by bonds between R-groups.	It is held by hydrogen bonds between backbone C=O and N-H groups.
Disulfide bonds are hydrogen bonds.	Disulfide bonds are strong covalent bonds between two cysteines.
All proteins have quaternary structure.	Only proteins with two or more polypeptides do.
Insulin is a conjugated protein.	Insulin has no prosthetic group; haemoglobin does.
Denaturation breaks peptide bonds.	Denaturation breaks the bonds holding the 3D shape; the sequence is unchanged.

Check that you can...

1. **Classify** R-groups, and **explain** how they create protein diversity
2. **Explain** how the primary structure determines a protein's shape
3. **Describe** alpha helices and beta-pleated sheets and the bonds that hold them
4. **Outline** four kinds of bond in tertiary structure, and how pH and heat affect them
5. **Predict** where hydrophilic and hydrophobic amino acids lie in soluble and membrane proteins
6. **Compare** quaternary structure in insulin, collagen and haemoglobin
7. **Relate** the form of globular and fibrous proteins to their functions

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Name the type of bond that stabilizes an alpha helix, and the atoms it joins.
2. State where non-polar amino acids are found in a soluble globular protein.
3. Explain why haemoglobin is called a conjugated protein.

Image credits

1GZX Haemoglobin.png – Zephyris at English Wikipedia – CC BY-SA 3.0

AminoAcidball.svg – GYassineMrabet – Public domain

Main protein structure levels en.svg – LadyofHats – Public domain

Both helix and the beta-sheet structures are held by specific hydrogen-bonding interactions.png – Unknown author – CC BY-SA 4.0

Insulin hexamer 4INS.png – Fvasconcellos (talk · contribs) – Public domain

Collagentriplehelix.png – Author not stated – CC BY-SA 3.0

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